

ARTICLE TYPE

Data-Driven Analysis of Musical Form and Harmonic Structure in AI-Generated Popular Music: A Case Study with Suno and Udio

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Abstract

We present a large-scale comparison of the formal and harmonic structures of music generated by users of two AI music generation platforms, Suno and Udio, against human-composed popular music. We address three questions: 1) How does AI-generated music's harmony relate to human popular music? 2) What structural similarities or differences exist? 3) What considerations arise when applying music information retrieval (MIR) tools to AI-generated music? We analyze 20,002 Suno tracks, 20,002 Udio tracks, and 20,010 human-composed recordings. Harmonic analysis uses a state-of-the-art conformer-based chord estimation model; structural analysis uses a joint beat-tracking and semantic section labeling model (All-In-One). We model chord progressions as trigrams and apply byte-pair encoding to section-label sequences to quantify formal diversity across the three collections. Analysis of over 2.52 million chord trigrams reveals that Suno concentrates its harmony on Western pop patterns such as I--V--vi--IV, while Udio exhibits more balanced chord distributions resembling human-composed music. Byte-pair encoding analysis shows Suno tends toward simple verse-chorus repetitions, whereas Udio exhibits the widest structural variety. These findings suggest that Suno (versions 3.5 and below) converges on a narrow harmonic and formal space, whereas Udio exhibits tendencies closer to human-composed music. We release beat-level chord annotations, metadata, section boundaries, self-similarity matrices, audio links, and code under an MIT license.

Keywords: AI Music, Generative AI, Musical Form, Harmonic Structure, Automatic Chord Estimation, Exploratory Data Analysis

1. Introduction

Commercial AI music generation platforms are becoming popular, and the resulting music is beginning to flood streaming services (Sturm et al., 2024). Suno,¹ launched online in late 2023, claimed in November 2024 to have a user base of 25 million people.² Udio,³ launched in the Spring of 2024, reported soon after that its users were generating ten songs a second.⁴ The summer of 2025 witnessed the appearance of an AI-generated music album by the fake “indie rock band” The Velvet Sundown, which has received millions of streams on Spotify.⁵ The French music streaming service Deezer announced in September 2025 that they are receiving 30,000 “fully AI-generated tracks daily”.⁶

This growing catalog of AI-generated music raises

several interesting questions for musicology and presents challenges for music information retrieval (MIR). How does the harmonic content of AI-generated music relate to the harmonic practices found in human-composed music? What structural similarities and differences are there between AI-generated and human-composed music? Given the rate at which this catalog is growing, how reliable and scalable are the computational methods of MIR for addressing these questions?

In this article, we address these questions specifically in the context of AI-generated music, drawing on examples from Suno and Udio. We describe the creation of a large corpus of 60,014 music recordings, balanced across three sources: 20,002 songs from Suno users, 20,002 songs from Udio users, and 20,010 human-composed Western popular songs.

Recent advances in MIR have made corpus-scale harmonic and structural analysis of popular music in-

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creasingly viable. Even though the challenges of resolving chord and structural information across diverse musical types remain an open problem, we employ consonance-ACE (Poltronieri et al., 2025) for harmonic analysis and All-In-One (Kim and Nam, 2023) for structural analysis, two recent implementations that bring the analysis significantly closer to musicological utility. The results of our harmonic analyses reveal that music from Suno exhibits strong stylistic characteristics of Western pop chord progressions, whereas music from Udio exhibits more balanced chord-quality distributions, resembling those found in human-composed music. The results of our structural analyses show that music from Suno tends toward simple AB (verse-chorus) repetition, whereas music from Udio exhibits a wider range of structural variations.

In summary, our contributions are: (i) a large comparative analysis of harmonic and formal properties in human-made and AI-generated music using a consistent pipeline, (ii) publicly released annotations for 60,014 tracks, including segmentations, beat-level chord estimates, style tags (Oramas et al., 2018), and audio metadata, and (iii) open-sourced code to reproduce the experiments.⁷ The remainder is organized as follows. The next section reviews related work. Section 3 describes the corpora used in the study. Section 4 motivates and describes our pipeline for harmonic and formal analysis. Section 5 presents our results. Finally, Sec. 6 discusses implications of our work, its limitations, and future directions.

2. Background

The computational analysis of musical structure and harmony is a focus of MIR (Müller, 2015).

2.1 AI Music Generation

Recent open-source work on text-conditioned music and audio synthesis includes models such as MusicGen (Copet et al., 2023), MusicLM (Agostinelli et al., 2023), and Stable Audio Open (Evans et al., 2025), representing the state of the art in publicly available generative audio research. However, the focus of this paper is not on the generative models themselves, but on the musical properties of their output, analyzed at the corpus scale in comparison with human-produced music.

2.1.1 Evaluation of AI-Generated Music

Sturm and Ben-Tal (2017) evaluates folk-rnn, a generative system trained on a large corpus of folk tunes in ABC notation, examining outputs in relation to musical practice rather than only automatic metrics, with a concrete evaluation pipeline, corpus statistics analysis, and focused musical inspection to locate where model outputs resemble and where they diverge from human repertoire. Ariza (2009) shows that interrogator-style authorship discrimination tests only establish that a digital music artifact sounds plausible to listeners, which is insufficient to characterize the musical quality

produced by an AI system.

Dervakos et al. (2021) propose a framework for evaluating symbolic AI-generated music using tools from music theory, defining heuristic metrics grounded in tonic coordinate systems and tone networks, such as the circle of fifths, to quantify musicality without relying on user surveys or real data.

Wu and Yang (2020) analyzes the outputs of the Jazz Transformer using objective metrics, including pitch-class histogram entropy, chord-progression irregularity, groove-pattern similarity, and structureness indicators, demonstrating that systematic shortcomings in AI-composed music are not fully captured by listening tests alone. Notably, their definition of chord progression irregularity as the proportion of unique chord trigrams in a piece establishes a precedent for trigram-based harmonic analysis of AI-generated music.

Some studies find biased perception of music when works are labeled as AI (Ragot et al., 2020; Shank et al., 2022; Déguernel and Sturm, 2023), others report null effects (Pasquier et al., 2016; Zlatkov et al., 2023). Together, these motivate a careful, data-driven comparison between music made with current AI platforms and music created by humans.

2.2 Music Structure Analysis

Music Structure Analysis (MSA) is the task of segmenting an audio recording into contiguous, non-overlapping sections and assigning labels that reflect musical function or similarity (Nieto et al., 2020). Computational approaches exploit four main principles: homogeneity (sections internally consistent with respect to some musical attribute), repetition (detecting recurrent sequences), novelty (identifying boundary points where musical content changes abruptly), and regularity (statistical properties of segment durations). Classical methods relied on self-similarity matrices and novelty curves derived from hand-crafted features such as chromagrams and MFCCs (Foote, 2000), repetition and recurrence-based representations (Levy and Sandler, 2008; Serrà et al., 2012), Laplacian spectral decomposition (McFee and Ellis, 2014), and barwise self-similarity with dynamic programming (Marmoret et al., 2023). Despite these advances, MSA remains an open challenge due to the subjectivity of structural annotations: the best reported structure labeling scores reach approximately 74% on the Harmonix Set (Kim and Nam, 2023), well below the estimated human inter-rater agreement of approximately 90% (Nieto et al., 2020).

More recent approaches have improved structural analysis through deep learning, including few-shot embeddings for Laplacian decomposition (Salamon et al., 2021), unsupervised contrastive multi-level representations (Buisson et al., 2022), self-supervised novelty detection (Peeters, 2023), hierarchical spectral-temporal Transformers for direct semantic labeling (Wang et al., 2022), and graph attention networks for pairwise struc-

ture prediction (Buisson et al., 2024).

For functional structure extraction, we employ the All-In-One model (Kim and Nam, 2023), which jointly performs beat tracking, downbeat detection, segment boundary detection, and semantic section labeling (verse, chorus, bridge, etc.) in a single architecture. We selected it because it directly outputs the section labels our analysis requires and achieves state-of-the-art performance on the Harmonix Set across all four tasks.

2.3 Automatic Chord Estimation

Automatic Chord Estimation (ACE) is a central task in MIR with applications in music performance, transcription, and formal analysis research. ACE has evolved from early template and chroma-based systems with probabilistic sequence models, such as Hidden Markov Models, Conditional Random Fields, or Deep Belief Networks (Mauch and Dixon, 2009; Ni et al., 2012; Humphrey and Bello, 2015), to more recent deep learning approaches that learn end-to-end representations from spectral features, increasingly targeting large chord vocabularies that include sevenths, extensions, and inversions.

Large-vocabulary ACE has advanced through several complementary strategies. McFee and Bello (2017) decouple pitch-class detection from label assignment in a 170-class framework; Jiang et al. (2019) decompose chord labels into independent musical components (root, quality, bass, extensions), generalizing to 301+ classes; and Park et al. (2019) propose BTC, a bi-directional Transformer achieving competitive large-vocabulary performance with publicly available code. Most recently, Poltronieri et al. (2025) propose *consonance-ACE*, a Conformer-based model whose decomposed decoder reconstructs chord labels without a fixed vocabulary, paired with consonance-based label smoothing, outperforming BTC on both standard and non-binary evaluation metrics on RWC Pop and USPop. Waseem Akram et al. (2025) further extends this line with ChordFormer for large-vocabulary recognition.

Despite these advances, accessible implementations remain limited in coverage. Open-source libraries such as Madmom (Böck et al., 2016) and librosa (McFee et al., 2025) provide accessible implementations of baseline chord estimation algorithms, most of which remain fundamentally limited to detecting major and minor triads, struggling with tetrads, extensions, suspensions, and inversions. Earlier efforts to expand chord vocabularies, such as Chordino (Mauch and Dixon, 2010), while still functional as a Vamp plugin, suffer from platform- and dependency-compatibility issues; the library is no longer maintained and cannot be integrated into modern Python pipelines. Furthermore, analyses relying on chord-class aggregation can misrepresent stylistic progressions: Collins (2025) demonstrate that Chordino-based histograms failed to capture modal-

interchange progressions while detecting only broad palette differences. Multi-source approaches such as DECIBEL (Odekerken et al., 2020), which fuse audio features with crowd-sourced MIDI annotations and guitar tablature, partially address vocabulary restrictions, yielding improvements of 0.5–13.6 percentage points over audio-only baselines on 180 Beatles songs and 20 Queen songs from the Isophonics chord-annotation corpus⁸, though major and minor accuracy typically remains between 75% and 87%, a ceiling attributed to annotation subjectivity.

Roman Numeral Analysis (RNA) encodes chords relative to a tonal center, but benchmarks on the PARC dataset (Poppe et al., 2025) confirm that even state-of-the-art models achieve modest accuracy on real polyphonic audio, particularly for complex progressions.

Given these considerations, this study adopts *consonance-ACE* (Poltronieri et al., 2025) for three reasons: (1) its decomposed decoder is not constrained to a fixed vocabulary, accommodating harmonic diversity beyond major/minor triads; (2) consonance-based label smoothing produces errors that fall between harmonically related chords rather than random ones; and (3) its demonstrated superiority over BTC on standard benchmarks, combined with publicly available code, provides a reproducible foundation for large-scale analysis. The model is described in detail in Section 4.1.

2.4 Harmony Analysis of Corpora

Several annotated corpora and computational analyses have contributed to the study of harmony and musical structure in popular music. A hand-annotated corpus of rock songs provides detailed harmonic and melodic analysis, enabling statistical investigation of chords, scale degrees, and key-finding models (Temperley and Clercq, 2013). Expert ground-truth chord annotations for charting songs support research in automatic chord recognition and music analysis (Burgoyne et al., 2011). Computational analysis of Billboard chart data has been used to study the evolution of harmony, timbre, and style in popular music from 1960 to 2010 (Mauch et al., 2015). Long-term trends in harmonic and timbral features of contemporary Western popular music have also been examined to document changes in musical diversity (Serrà et al., 2012). Automated approaches to chord and key estimation provide frameworks for the simultaneous identification of chords and musical context from audio data (Mauch and Dixon, 2009). However, there is no comprehensive corpus-level comparison of harmonic and structural properties between current AI platforms and human music.

3. Corpora

The corpora are formed as a collection of three audio datasets (summarized in Table 1) containing audio files from Suno and Udio, from the AI-generated side, and human-generated music found on Lastfm cat-

alog. The audio tracks originate from (Cros Vila et al., 2025), where web scraping was performed for the Suno and Udio datasets between May and October 2024. Suno data was collected from its frequently updated "New Songs" playlist⁹, queried every two hours to capture its daily influx of 50 new tracks. Udio data was collected from the platform's publicly accessible front page. Both datasets include metadata and audio files. The audio file folder contained more than 200,000 files, from which we randomly selected approximately 20,000 audio files from each dataset for analysis. The corpora uses the Million Song Dataset (MSD) (Bertin-Mahieux et al., 2011)¹⁰ to confirm the audio files are human-created music, leveraging Lastfm's associated YouTube URLs to download around 200,000 unique tracks, where we randomly chose 20,010.¹¹ We thus refer to this collection as "Lastfm".

Dataset	# JSON # Files	Duration Mean $\pm$ S.D.	Total Duration
Lastfm	20,010	261.3 $\pm$ 182.1s	1452h 6m 56s
Suno	20,002	175.2 $\pm$ 54.6s	1084h 47m 9s
Udio	20,002	143.2 $\pm$ 98.6s	801h 47m 36s

Table 1: The number of files is formed by a parallelized process with workers extracting form and chords data from a randomized folder of more than 100,000 audio files, with a limit of 20,000 JSON files

To address potential concerns about selection bias, we report user engagement metrics—likes and plays—across the Suno and Udio collections. Table 2 presents summary statistics, and Figure 1 displays their distributions on a logarithmic scale. Both platforms share identical median values (1 like and 4 plays), with means driven upward by a small number of high-engagement outliers. The 90th percentile reaches only 2 likes and 26 plays for Suno, and 3 likes and 32 plays for Udio, confirming that the vast majority of tracks in both collections received minimal engagement. A popularity-filtered dataset would yield a distinctive median difference relative to an unfiltered baseline; the equivalence observed here is consistent with unfiltered acquisition from each platform's front page. No ranking, curation, or engagement threshold was applied to either collection.

Metric	Collection	Mean	Med.	P90	P95	Max
Likes	Suno	5.25	1	2	4	34,921
	Udio	1.83	1	3	8	898
Plays	Suno	204.88	4	26	50	445,490
	Udio	44.10	4	32	58	48,940

Table 2: Summary statistics of likes and plays for the Suno ($n = 19,972$) and Udio ($n = 19,992$) collections. P90 and P95 denote the 90th and 95th percentiles.

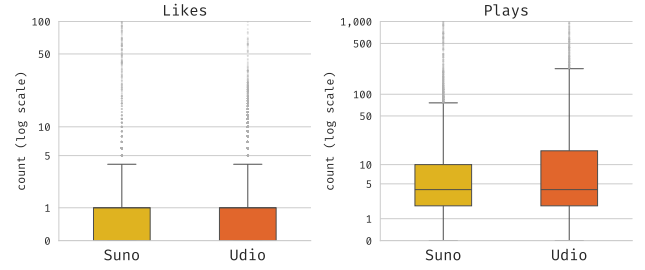


Figure 1: Box plots of likes and plays for the Suno and Udio collections on a logarithmic scale. Both platforms share a median of 1 like and 4 plays.

4. Method

This section describes the pipeline used to extract harmonic and structural information from the 60,014 audio recordings in our corpus, and the derived representations for comparative analysis.

4.1 Automatic Chord Estimation

Chord estimation is performed using `consonance-ACE` (Poltronieri et al., 2025), which independently predicts root, bass, and pitch-class activations and reconstructs chord labels symbolically through a decomposed decoder. The model operates on CQT spectrograms computed over 20-second chunks and outputs frame-level chord predictions with no constraint to a fixed vocabulary. We use the publicly available pretrained model without modification. Chord segments shorter than 0.5 seconds are merged into adjacent segments, and consecutive frames with identical chord labels are collapsed into continuous intervals.

The output of the chord estimator consists of absolute chord labels (e.g., D:min7, G:7, C:maj7) aligned with a time onset-offset columns as shown in Table 3. To enable key-relative analysis, we convert these to functional harmony representations (e.g., ii, V, I). For each track, we first estimate the global key by computing the pitch-class distribution from the chord-root sequence and correlating it with the major- and minor-key profiles (Krumhansl and Kessler, 1982). Each absolute chord label is then mapped to its corresponding Roman numeral degree relative to the estimated key using the `music21` library (Cuthbert and Ariza, 2010). Consecutive identical Roman numerals are deduplicated before downstream n-gram analysis. The resulting chord intervals are subsequently aligned to beat positions estimated by the All-In-One model (Section 4.2), yielding the beat-level chord annotations included in the released dataset.

4.2 Musical Form Estimation

Musical structure is extracted using the All-In-One model (Kim and Nam, 2023), which jointly performs four tasks in a single architecture: beat tracking (locating regular pulse positions), downbeat detection (identifying the first beat of each bar), segment boundary

Start (s)	End (s)	Chord
0.000	0.951	N
0.951	2.900	G:min
2.900	4.501	F:maj
4.501	7.471	D#:maj
7.471	9.118	G:min
9.118	11.346	F:maj
11.346	12.691	C:min
12.691	13.805	D#:maj
13.805	15.360	G:min
15.360	16.914	F:maj
16.914	18.840	D#:maj6
18.840	21.787	G:min
21.787	23.179	F:maj
23.179	24.617	D#:maj
24.617	25.499	G:min
25.499	26.241	D#:maj

Table 3: Example output of consonance-ACE. Each row contains the onset time, offset time, and estimated chord label. N denotes no chord detected.

detection (finding time points where sections change), and functional structure labeling (assigning semantic labels such as verse, chorus, bridge, intro, outro, or solo to each segment).

The model first separates the input audio into four stems (drums, bass, vocals, other) using Hybrid Transformer Demucs (Rouard et al., 2023) and processes the resulting spectrograms through a stack of transformer modules combining dilated and neighborhood attention. Four output heads produce frame-level predictions for each task. The model achieves state-of-the-art performance on the Harmonix Set (F1 = 0.958 for beats, F1 = 0.915 for downbeats, HR.5F = 0.660 for segmentation, PWF = 0.738 for structure labeling).

We apply the publicly available pretrained model without modification. For each track, the model outputs a sequence of labeled sections with their onset and offset times, stored as `_musical_form.lab` files. Consecutive identical section labels are subsequently merged before downstream analysis.

4.3 Trigram Dataset Generation

To systematically analyze harmonic tendencies across collections, we constructed a trigram dataset—a corpus of all consecutive three-chord sequences—extracted from each collection’s data. For example, the progression *Dm7, Ab7, Gm7, C#7, ...* yields [*Dm7, Ab7, Gm7*], [*Ab7, Gm7, C#7*], and so on. The process is as follows:

1. **Normalization:** Absolute chord labels are first converted to Roman numerals and consecutive identical Roman numerals are de-duplicated, as described in Section 4.1. The example above uses absolute chord labels for illustrative clarity. Sequences containing “no-chord” labels were not considered for trigrams.
2. **Trigram Generation:** For each song, we extracted overlapping chord trigrams from the Roman numeral sequence using a sliding window approach:

$$(C_i, C_{i+1}, C_{i+2}), \quad i = 1, \dots, N - 2$$

where N is the number of chord events in the song after de-duplication, and C_i denotes the i^{th} Roman numeral. All trigrams were retained, including repetitions and sequences spanning sections or boundaries, unless a song or section was shorter than three chords.

4.4 Byte-Pair Encoding of Harmonic and Formal Sequences

To compare structural and harmonic diversity across collections, we apply Byte-Pair Encoding (BPE) (Sennrich et al., 2016) to two types of sequences. BPE is a sub-word tokenization algorithm that iteratively merges the most frequent adjacent symbol pairs in a corpus, building a vocabulary of recurring patterns from the bottom up.

For musical form, each track’s section sequence is first converted into overlapping trigrams of consecutive section labels (e.g. *verse · chorus · verse*), and BPE is applied using each trigram as the basic unit. BPE is run for up to 80 merge operations per collection, which is sufficient to exhaust all meaningful merges given the small section-label vocabulary.

For harmonic sequences, we first extract chord trigrams from the Roman numeral sequences produced by the ACE pipeline (Section 4.1). A trigram is a window of three consecutive, mutually distinct Roman numeral symbols. Each track’s sequence is represented as a series of trigrams, and BPE is applied at the trigram level: each token fed to the algorithm is itself a trigram, and merge operations identify the most frequent pairs of adjacent trigrams. BPE is run for 150 merge operations per collection, an empirical cap chosen to capture the most frequent recurring patterns without over-merging sparse ones. The resulting merge history, restricted to base-to-base merges (i.e., pairs in which both tokens are single trigrams), reveals the most common harmonic progressions of four chords and their cross-collection frequency distributions.

4.5 Axis of Awesome Chord Progression Analysis

We measure how often songs in each collection contain chord progressions matching the Axis of Awesome (Richards, 2017), using the Roman numeral sequences produced by the ACE pipeline (Section 4.1). Four-chord sequences are formed as p-trigrams, pairs of adjacent chord trigrams, consistent with the trigram vocabulary used throughout this paper (Section 4.4).

The Axis of Awesome is associated with three four-chord sets: the major form {I, V, vi, IV}, the minor form {i, bVI, III, bVII}, and a minor variant {i, bVI, bIII, bVII}. We treat all 24 orderings of each set as valid match targets, giving 72 distinct four-chord patterns in total. This decision is deliberate: the analysis targets harmonic vocabulary membership, whether a track draws on axis-family pitch material, rather than the syntactic function of any specific ordering. As Richards (2017) discusses, rotations of the same four-chord set

can carry different tonal implications; the reported axis rates should therefore be read as a measure of axis-family prevalence, not as evidence of a single recurring functional progression. From here, we will refer to this pattern of chord progressions as Axis.

All Roman numeral labels are first normalized enharmonically by mapping each label to its semitone position and back (e.g., $\#V \rightarrow bVI$), and consecutive identical labels are removed. Trigrams are then extracted from each collapsed sequence using a sliding window of three chords; only trigrams in which all three chords are distinct are retained, consistent with the BPE trigram definition. Adjacent trigram pairs are formed from the retained list: a pair $(\langle a, b, c \rangle, \langle b, c, d \rangle)$ is valid only if the second trigram starts exactly one position after the first in the original sequence (i.e., they share a two-chord overlap). Each valid pair reconstructs a four-chord window (a, b, c, d) , which is tested for exact membership in the set of 72 axis patterns.

The p-trigram axis rate for a song is the proportion of valid p-trigram pairs whose reconstructed window matches any axis pattern, expressed as a percentage.

5. Comparing Patterns in Form and Harmony

This section is divided into four stages, moving from corpus-level harmony to evaluation of the analysis pipeline. Section 5.1 first quantifies how often each three-chord “trigram” occurs in Lastfm, Suno, and Udio, revealing collection-specific harmonic fingerprints. We then compare overall chord-quality vocabularies and highlight progressions that are disproportionately common in Suno (Section 5.2). Next, in Section 5.3, a Byte-Pair-Encoding (BPE) analysis of segment labels uncovers the most frequent formal patterns and contrasts the structural diversity of the three catalogs. Finally, in Section 5.4 we report the percentage of Axis chords found on each collection. Together, these four strands address our central question: To what extent do the harmonic tendencies of AI-generated music relate to those of human-composed works? What structural similarities or differences do we find when comparing AI-generated music against human-composed music?

5.1 Trigram Tendencies Across Collections

Figure 2 shows the thirty most prevalent trigram patterns in the three members of the collection (Lastfm, Suno, and Udio), distributed over the ten most common styles found by extracting the metadata descriptors of the songs. The label ‘Others’ recollects several styles in one group. The ‘Jazz’ annotation in Suno or Udio primarily applies to ambient instrumental music, with no particular harmonic patterns typical of standard jazz styles. Both Udio and Suno show a strong tendency to Pop, Rock, Metal, and Electronic.

While we acknowledge that the style tags associated with tracks in our dataset are inherently flawed and subject to user bias (users may arbitrarily assign

genres, and the platforms’ own tagging systems lack rigorous musicological grounding), we use these classifications as they reflect the styles users most frequently specify when generating music. As demonstrated by (Casini et al., 2025), these same style descriptors (Pop, Rock, Electronic, Hip-hop, etc.) are among the most prevalent tags in the dataset.

The collection-level differences visible in Figure 2 are quantified in detail in Sections 5.2–5.4.

In Figure 3, an equivalent distribution of chord qualities is seen in all three collections. The major chord is the most common, followed by minor, power chords, suspended fourths, and dominant sevenths. As an observation, dominant chords with sharp ninth or flat thirteenth, also altered dominant, are commonly used in minor and major modes, as a modal interchange in major tonalities. The dominant sharp eleventh is not common; the once-ubiquitous dominant substitution used in jazz, bossa-nova, or soul has become an endangered harmonic particularity in all three collections. Power chord, as a neutral quality, is used with mixed frequency across major and minor tonalities.

Figure 4 shows a distribution of the tonalities found in all collections. The quantification is sorted to Lastfm occurrence. The proportions show similarities on AI-generated music platforms; nevertheless, Suno shows stronger tendencies to produce C major, D major, G major, and B major, compared to Lastfm and Udio. Udio has a prominent presence of A minor, G minor and Eb minor. The distribution of tonalities is consistent with the cycle of the fifth.

Figure 5 reports the twenty most frequent chord trigrams found in the styles Rock, Electronic, Pop, Hip-Hop, Jazz (Chill-Out), Country, Blues, Soul, and Rhythm and Blues. Each collection has similarities and also strong tendencies revealing distinct harmonic signatures across Lastfm, Suno, and Udio. For example, in Soul, Udio and Suno are prone to harmonies based on dominant-subdominant permutations (V-I-IV, IV-V-I and I-IV-V).

5.2 Harmonic Tendencies

Figure 6 shows the relative frequency of canonical pop/rock chord trigrams — defined as permutations of $\{I/i, IV/iv, V/v, vi/VI/bVI\}$ — in each collection, computed as risk ratios against Lastfm and Udio using the method described in Section 4.3. The figure demonstrates that Suno tends to favor chord progressions strongly associated with the Western pop idiom, particularly those prevalent since the early 2000s. For every progression tested, Suno’s relative frequency is significantly greater than that observed in the other collections ($p < 0.001$ for all comparisons). As shown in (Temperley and Clercq, 2013), the centrality of progressions such as $I--V--vi--IV$, $IV--I--V$, and related idioms is a defining feature of Western pop/rock music; they report that the most frequent non-tonic chord roots are IV , V , $bVII$, and vi , with progressions among

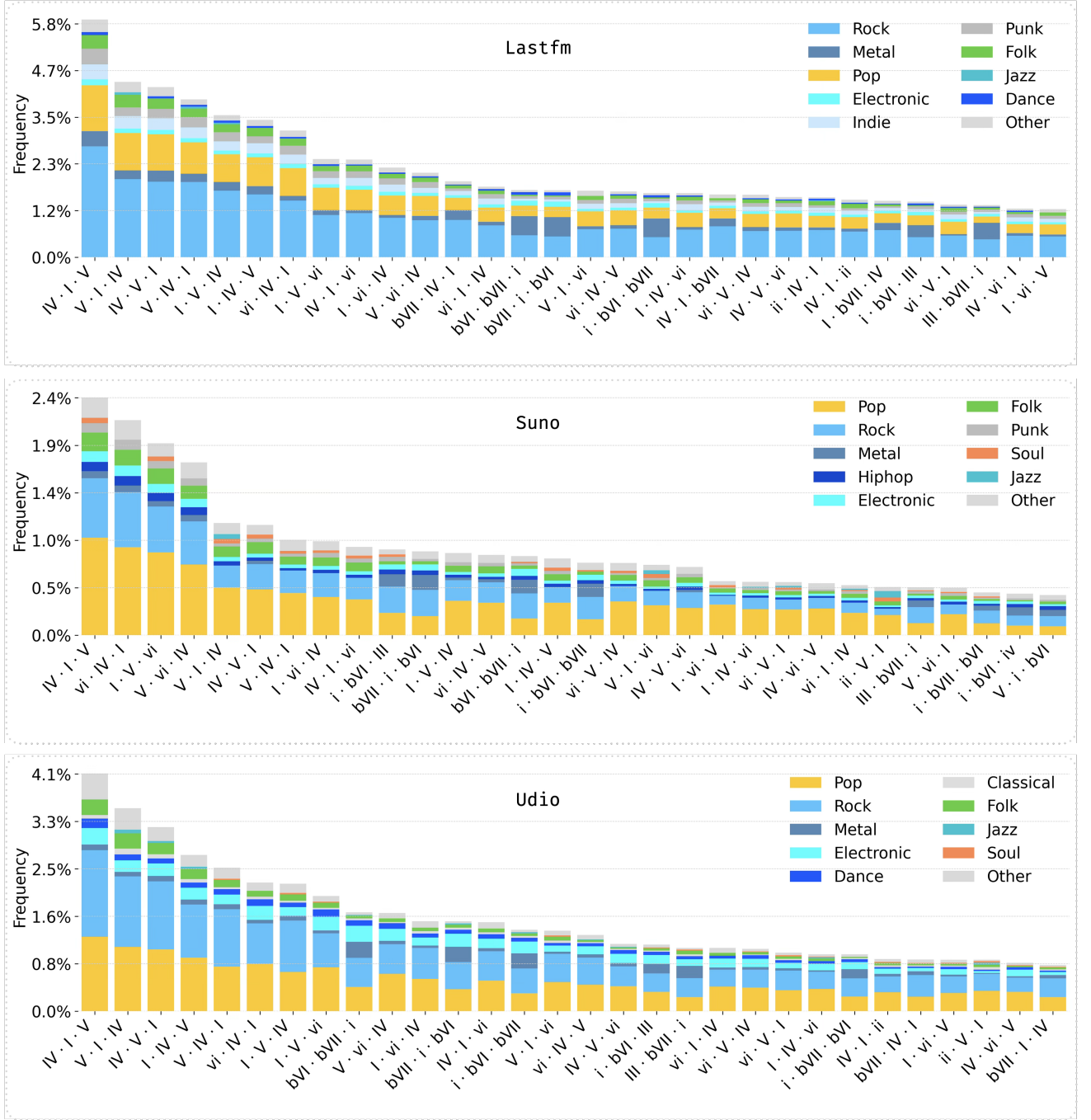


Figure 2: Top 30 most common trigrams found in Lastfm, Suno, and Udio datasets. The y-axis shows the relative frequency of each trigram in the collection. The distribution of styles differs across collections.

these chords forming the core stylistic harmonic vocabulary. Our results both replicate and extend these findings, revealing the extent to which such progressions are even more prevalent in AI-generated music from Suno.

To quantify the tendencies shown in Figure. 5 for each canonical progression in Suno relative to the reference collections (Lastfm and Udio), we calculated a risk ratio (RR) for every trigram. Raw counts are misleading because Suno contains far more trigrams than the reference corpus, so the RR normalizes by each col-

lection’s total:

$$RR(t) = \frac{N_{\text{Suno},t}/N_{\text{Suno}}}{N_{\text{Ref},t}/N_{\text{Ref}}}$$

where $N_{\text{Suno},t}$ and $N_{\text{Ref},t}$ are the counts of trigram t in Suno and in the reference corpus, respectively, while N_{Suno} and N_{Ref} denote the total numbers of trigrams in those two corpora.

To test whether the difference is greater than expected by chance, we build a 2×2 contingency table (Suno vs. Reference $\times t$ vs. “all other trigrams”) and

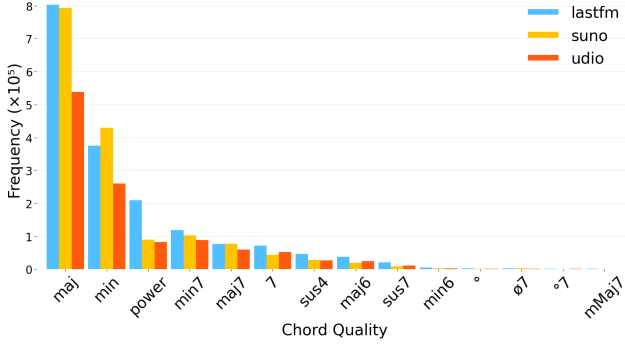


Figure 3: Vocabulary distribution of chord qualities over the collections. The y-axis indicates the total number of occurrences for each chord quality in each collection, scaled by $\times 10^5$.

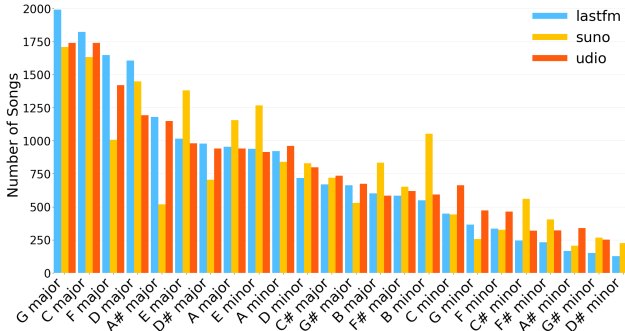


Figure 4: Distribution of most common tonalities in all three collections.

apply Fisher’s exact test. All raw p -values are corrected for multiple comparisons with the Benjamini–Hochberg false-discovery-rate procedure (Benjamini and Hochberg, 1995); the resulting p -value is used to filter and choose the trigrams that are shown in the Table 4:

- at least 30 occurrences in Suno and 10 in the reference,
- risk ratio $RR \geq 2$,
- adjusted p -value < 0.01 .

	Target trigram	Other trigrams
Suno	a	b
Reference	c	d

where $a = N_{\text{Suno},t}$, $b = N_{\text{Suno}} - a$, $c = N_{\text{Ref},t}$, and $d = N_{\text{Ref}} - c$. A risk ratio $RR_t > 1$ together with $p < 0.01$ indicates that trigram t is significantly more common in Suno than in the reference corpus, as is the case for all the trigrams shown.

Table 4 shows a representative selection of the 118 Suno-enriched trigrams passing all filters: Panel A lists the top 15 by Suno count of the 53 canonical progressions (permutations of $\{I/i, IV/iv, V/v, vi/VI/bVI\}$); Panel B lists the top 10 non-canonical trigrams by risk ratio from the remaining 65. The columns provide the raw counts in Suno (C_{Suno}), Udio (C_{Udio}), and Lastfm

(C_{Last}), their percentage of the total vocabulary, and the risk ratio relative to Lastfm.

Across all canonical progressions in Panel A, risk ratios range from $2.1\times$ to $3.3\times$, with the major-mode axis chains (e.g., $I--V--vi$ at $3.2\times$, $vi--IV--I$ at $2.7\times$) among the most enriched. Every canonical pattern is significantly enriched ($p\text{-adj} < 10^{-16}$).

Trigram	C_{Suno}	C_{Udio}	C_{Last}	% Suno	% Udio	% Last	RR
Panel A: Top 15 canonical enriched trigrams							
vi, IV, I	34095	6442	8659	4.326	1.770	1.596	2.71
V, vi, IV	29383	6311	7311	3.728	1.734	1.347	2.77
I, V, vi	22803	3967	4948	2.893	1.090	0.912	3.17
V, i, bVI	4823	1221	1077	0.612	0.336	0.198	3.08
v, i, bVI	4610	863	1022	0.585	0.237	0.188	3.11
iv, V, i	4499	1373	1029	0.571	0.377	0.190	3.01
bVI, V, i	3508	1092	1074	0.445	0.300	0.198	2.25
iv, v, i	3287	653	739	0.417	0.179	0.136	3.06
V, i, iv	3273	1280	950	0.415	0.352	0.175	2.37
bVI, v, i	3117	618	658	0.396	0.170	0.121	3.26
i, v, bVI	2563	594	650	0.325	0.163	0.120	2.72
v, i, iv	2553	650	762	0.324	0.179	0.140	2.31
i, iv, V	2128	803	576	0.270	0.221	0.106	2.54
i, bVI, v	2081	554	555	0.264	0.152	0.102	2.58
i, iv, v	1929	581	622	0.245	0.160	0.115	2.14
Panel B: Top 10 non-canonical enriched trigrams (by risk ratio)							
iv, ii, v	93	17	10	0.012	0.005	0.002	6.40
iii, vi, ii	1485	236	181	0.188	0.065	0.033	5.65
bVI, iv, ii	74	25	10	0.009	0.007	0.002	5.10
II, v, III	245	30	35	0.031	0.008	0.006	4.82
IV, iii, vi	1390	276	199	0.176	0.076	0.037	4.81
V, biv, vi	198	33	30	0.025	0.009	0.006	4.54
IV, biv, vi	131	29	21	0.017	0.008	0.004	4.30
biv, vi, IV	195	39	32	0.025	0.011	0.006	4.20
iii, IV, v	143	26	24	0.018	0.007	0.004	4.10
iii, ii, v	67	19	12	0.009	0.005	0.002	3.84

Table 4: Suno-enriched trigrams. Panel A lists the most prevalent canonical progressions, defined as any permutation of chords drawn exclusively from $\{I/i, IV/iv, V/v, vi/VI/bVI\}$. Panel B lists the highest risk-ratio non-canonical trigrams. C = raw occurrence count; RR = risk ratio (Suno / Lastfm). All rows are statistically significant (FDR-adjusted $p < 0.001$).

5.3 Byte Pair Encoding Analysis of Segment Sequences

To explore common patterns in how songs are structured, we used Byte Pair Encoding (BPE) on sequences of labeled segments. Each sequence begins with `<init>` and ends with `<end>` to mark the song’s boundaries.

We applied BPE across the three music collections: 859,882 trigrams in Lastfm, 1,052,255 trigrams in Suno, and 608,663 trigrams in Udio; in total, 2,520,800 trigrams. This process finds and merges the most frequent adjacent tokens, revealing larger structural motifs that reflect how songs are typically organized.

Figure 7 shows the most frequent BPE merge pairs; the results are discussed in Section 6.3.

5.3.1 Form Diversity and Style Differences

We also computed the “Unique Tokens Ratio,” which reflects how many different segment types appear in a song. Lower values suggest repetitive forms; higher values indicate more varied structures. This metric was lowest in Suno (0.44 ± 0.18) and highest in Udio (0.58 ± 0.20), with Lastfm (0.51 ± 0.19) in between.

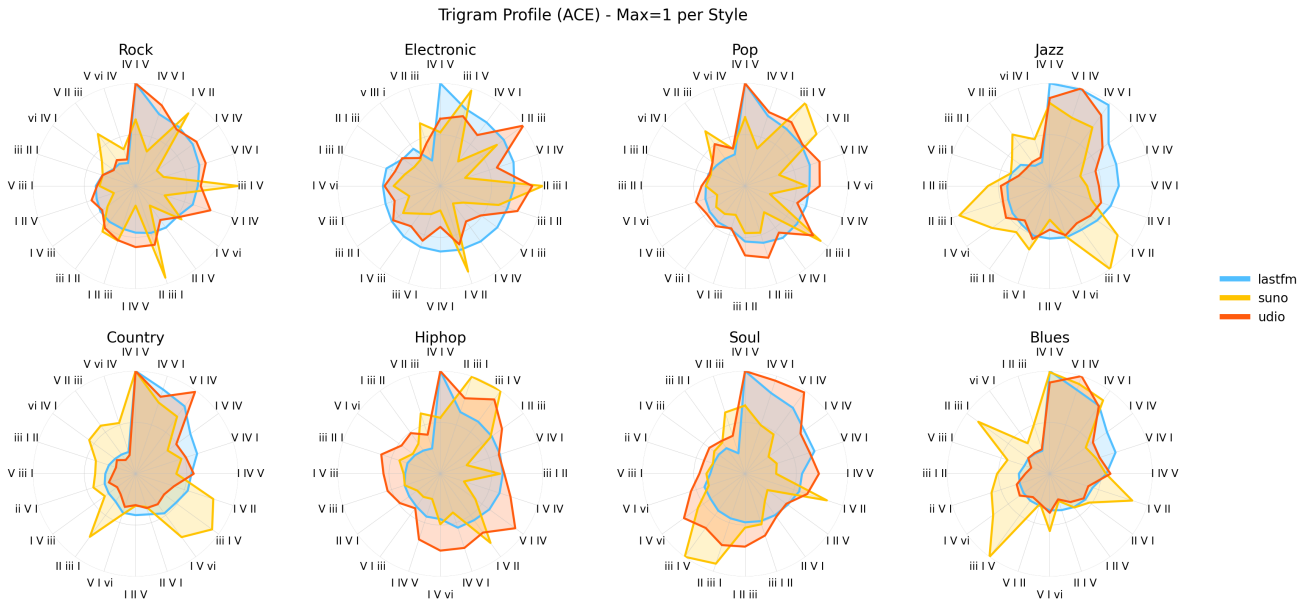


Figure 5: Spider-chart shows the common trigrams found per style in all collections. The spider chart is sorted by relevance, with Lastfm as the reference point.

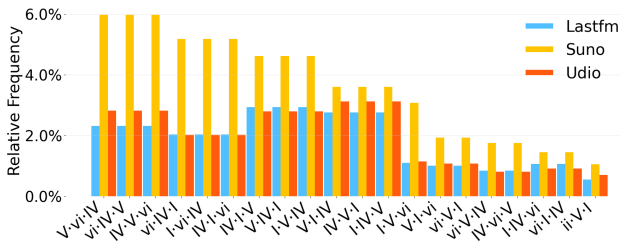


Figure 6: Relative frequency of canonical pop/rock trigrams in Suno compared with Lastfm and Udio. Bars show proportions of risk ratios, and all Suno-reference differences are significant at $p < 0.01$.

may therefore reflect not only a bias toward formulaic composition but also a constraint imposed by shorter generation lengths, as shown in Figure 10.

5.4 Axis of Awesome Tendencies

Table 5 reports the number and proportion of songs in each collection exceeding each p-trigram axis rate threshold.

Table 5: Proportion of songs per collection at each p-trigram axis rate threshold. The axis rate is the fraction of a song’s four-chord windows that match any rotation of the Axis of Awesome progression. A song at $\geq 50\%$ has more than half its harmonic windows following the Axis loop; $\geq 10\%$ indicates the progression is a structural element of the song. Collections: Lastfm ($n = 18,916$), Suno ($n = 18,978$), Udio ($n = 19,031$).

Thresh.	Lastfm		Suno		Udio	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
>0%	5,034	26.6	8,864	46.7	5,066	26.6
$\geq 1\%$	4,950	26.2	8,809	46.4	5,023	26.4
$\geq 5\%$	3,355	17.7	7,225	38.1	3,719	19.5
$\geq 10\%$	2,392	12.6	6,076	32.0	2,791	14.7
$\geq 25\%$	1,195	6.3	3,987	21.0	1,431	7.5
$\geq 50\%$	434	2.3	1,993	10.5	562	3.0

At the $\geq 10\%$ threshold, 12.6% of Lastfm songs, 14.7% of Udio songs, and 32.0% of Suno songs qualify as axis-based, confirming the findings of Richards (2017) on the cultural ubiquity of this progression and extending them to show that Suno reproduces it at a substantially higher rate relative to both the human reference corpus and Udio.

5.3.2 Summary

The results from our BPE analysis suggest several insights relevant to broader questions about the formal structures and creative capacities of AI-generated music.

First, we observe that Suno exhibits a strong preference for simple, repetitive structures, particularly repeated verse-chorus cycles. This convergence toward a reduced formal vocabulary likely reflects biases in generation objectives optimised for familiarity and accessibility, as well as the short song snippets to be extended later via the web interface. Udio shows similar structural tendencies but with merge frequencies closer to Lastfm. A key confound for both platforms is song duration: Suno tracks average 175 seconds (approximately 2.9 minutes), Udio tracks average 143 seconds (approximately 2.4 minutes), and Lastfm tracks average 261 seconds (approximately 4.4 minutes), making AI-generated tracks roughly half to two-thirds the length of their human-composed counterparts. The reduced structural diversity observed in Suno and Udio

BPE Trigram Pair	verse-chorus-verse → chorus	9621	11320	6749
	chorus-inst-verse → chorus	2757	209	299
	verse-chorus-bridge → chorus	2272	1291	918
	verse-chorus-solo → chorus	1963	970	833
	verse-chorus-solo → verse	1738	424	772
	verse-chorus-inst → chorus	1166	1193	840
	verse-chorus-solo → outro	846	253	305
	verse-inst-verse → chorus	612	438	504
	intro-chorus-verse → chorus	594	0	277
	inst-chorus-inst → chorus	548	234	282
	intro-verse-chorus → outro	449	391	244
	verse-chorus-inst → solo	444	0	204
	chorus-bridge-verse → chorus	427	0	0
	verse-solo-verse → chorus	400	0	241
	verse-inst-verse → inst	0	0	182
	chorus-verse-chorus → outro	0	271	0
	intro-inst-solo → outro	0	238	0
	verse-chorus-inst → verse	0	2103	1599
	inst-chorus-verse → chorus	0	0	193
	verse-chorus-verse → inst	0	251	231
	verse-chorus-verse → solo	0	0	199
	intro-inst-verse → chorus	0	0	177
	verse-chorus-inst → outro	0	209	0
		Lastfm	Suno Collection	Udio

Figure 7: Byte-Pair Encoding (BPE) merge-pair heatmap across collections. Each row represents a frequently merged pair of consecutive musical-form trigrams (trigram A → last cell of trigram B, as the first two cells are redundant). Cell shading encodes relative frequency within each collection; integer annotations report absolute merge counts. Pairs are sorted by descending Lastfm frequency.

6. Discussion

This study addresses three research questions about AI-generated music at corpus scale:

RQ1 How reliably can current MIR tools analyze harmonic structure, vocabulary, and musical form in AI-generated music?

RQ2 To what extent does the harmony of AI-generated music relate to human-composed music?

RQ3 What structural similarities or differences emerge when comparing AI-generated and human-composed music?

We discuss each in turn, then draw out two cross-cutting implications: outcome homogenization and creative agency (Section 6.4).

6.1 RQ1: Applying MIR Tools to AI-Generated Music

The chord estimator and structural model adopted in this study (Sections 4.1 and 4.2) address the princi-

pal methodological concerns for corpus-scale analysis, but challenges remain that should be acknowledged. For chord estimation, even state-of-the-art models exhibit lower accuracy on complex chord types (sevenths, extensions, suspensions, inversions) than on triads (Pauwels and Peeters, 2013), and reported performance is sensitive to evaluation vocabulary and chord mapping strategies. Annotations from the Polyphonic Audio to Roman Corpus confirm that current RNA models achieve modest accuracy on real polyphonic audio, especially for harmonically complex progressions (Poppe et al., 2025). In our pipeline, the conversion from absolute chord labels to Roman numerals depends on reliable global key estimation via pitch-class correlation against Krumhansl-Kessler profiles. While this method is standard, key estimation errors propagate to every downstream Roman numeral label in a given track. For tracks with key ambiguity, modulation, or modal mixture, the functional analysis may misrepresent the intended harmonic relationships. Music21 library also required normalization of chord names to form the functional related chord. Furthermore, consonance-ACE assumes enharmonic equivalence: sharp ninths, sharp elevenths, and flat thirteenthths are encoded as their enharmonic equivalents (flat third, flat fifth, and flat sixth respectively), and therefore do not appear as distinct labels in the chord vocabulary.

This study operates at the level of chord-label distributions and n-gram sequential patterns, which is the appropriate granularity for corpus-scale comparison across 60,014 tracks. Voice leading, harmonic rhythm, and chord-voicing detail are not captured in any chord-symbol-based corpus study and fall outside the present scope.

For structural analysis, the All-In-One model provides semantic section labels (verse, chorus, bridge, intro, outro) unavailable from unsupervised methods. Its reported scores on the Harmonix Set (HR.5F = 0.660 for segmentation, PWF = 0.738 for structure labeling) remain below estimated human inter-rater agreement (Nieto et al., 2020), so a degree of error is present in the annotations. However, since the same model is applied uniformly across all three collections, these errors affect each corpus equally. Our analysis targets distributional tendencies across 60,014 tracks; systematic differences between collections therefore reflect genuine structural tendencies rather than tool-specific artifacts.

Despite these caveats, the combination of a chord estimator with a semantically aware structure analyzer represents a meaningful advance over the triad-level, label-free pipelines that were the practical default for large-scale audio analysis until recently. The rapid development of both ACE (Poltronieri et al., 2025; Park et al., 2019; McFee and Bello, 2017; Jiang et al., 2019; Waseem Akram et al., 2025) and MSA (Kim and Nam,

Musical Form Trigram Profile — Human-Annotated Styles (Normalized)

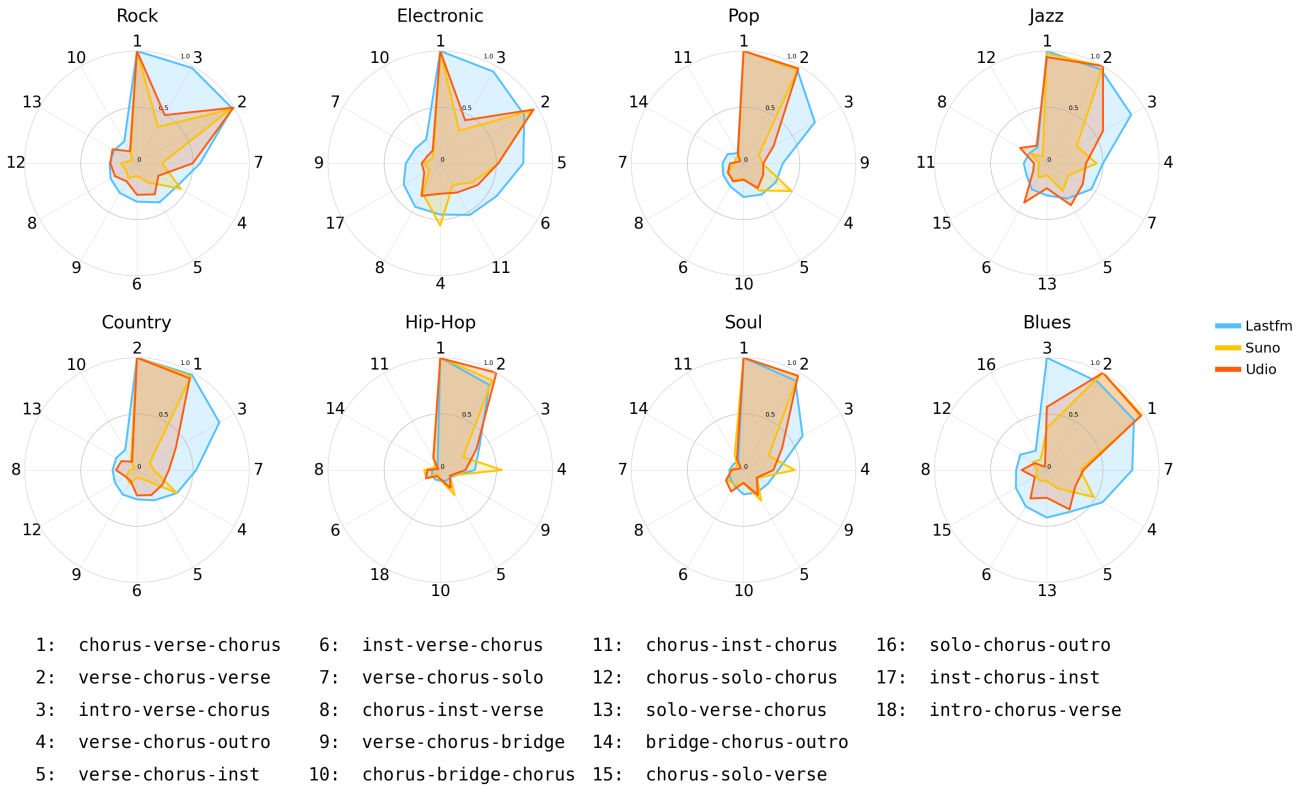


Figure 8: Normalized musical form trigram profiles across eight human-annotated genre tags per collection. Each axis represents a trigram of three consecutive section labels (e.g., chorus-verse-chorus), extracted after collapsing repeated adjacent sections. For each genre, the top trigrams are selected by Lastfm frequency, and values are normalized to the maximum count per collection within each genre.

2023; Wang et al., 2022; Buisson et al., 2024; Peeters, 2023) tools, with publicly released code and pretrained models, makes corpus-scale harmonic and formal analysis of this kind increasingly feasible.

6.2 RQ2: Harmonic Relationship Between AI-Generated and Human-Composed Music

The four metrics used in this section capture different dimensions of harmonic practice and should be read accordingly. Chord-quality distributions (Figure 3) describe the harmonic palette, which chord types appear and how often. Trigram frequencies and cross-collection correlations describe local sequential behavior, in which three-chord successions recur and at what relative rates. Axis rates measure membership in a canonical four-chord family, not the prevalence of any single ordering. BPE merge histories describe recurring, longer patterns, across which tetragrams and extended sequences are consolidated within a collection. These dimensions are complementary but independent: a high trigram-frequency correlation between two collections, such as the $r = 0.952$ shared by Udio and Lastfm, indicates distributional similarity in local harmonic motion but says nothing about whether those

progressions occupy the same formal positions within a song or occur at comparable rates of harmonic change.

Three strands of evidence converge on a consistent picture of how AI-generated harmony relates to human-composed music.

At the level of chord-quality vocabulary, the three collections are broadly similar: major triads dominate, followed by minor triads, power chords, seventh chords, dominant sevenths, and suspended chords (Figure 3). This indicates that current generative platforms reproduce the surface-level harmonic palette of popular music with fidelity. The differences emerge not in what chord types appear, but in how they are sequenced.

Trigram frequency analysis reveals that Suno concentrates its harmonic motion on a narrow subset of Western pop progressions. Across all 53 canonical progressions defined as permutations of $\{I/i, IV/iv, V/v, vi/VI/bVI\}$ that pass the enrichment filters (FDR-adjusted $p \ll 0.001$, $RR \geq 1.3$), Suno is overrepresented at frequency ratios ranging from $1.3\times$ to $3.3\times$ relative to Lastfm. The highest-frequency 15 of these 53 progressions, listed in Panel A of Table 4, concentrate the overrepresentation at $2.1\times$ – $3.3\times$, while the remaining lower-

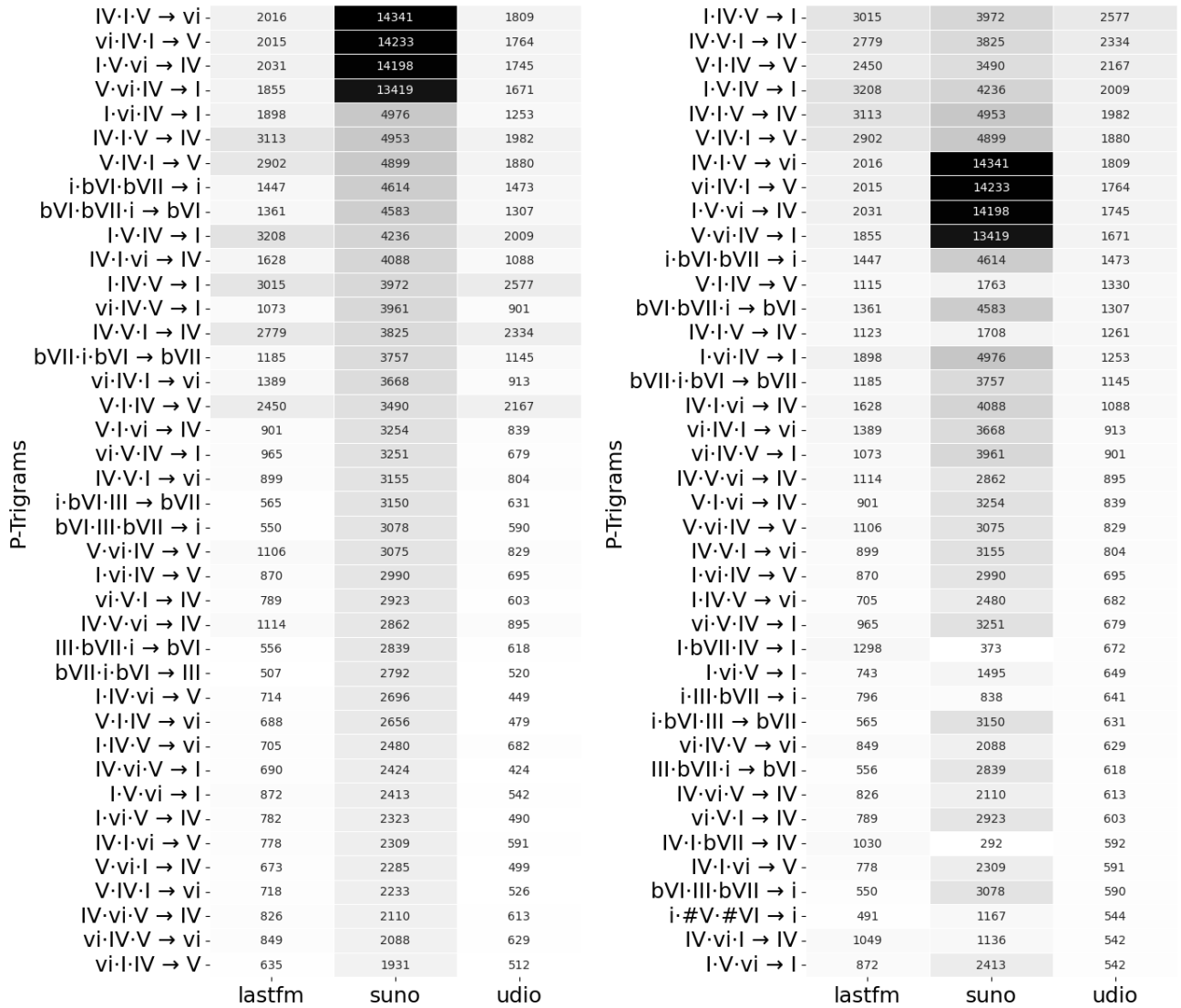


Figure 9: Byte-Pair Encoding treating the trigrams as cells. Two consecutive trigrams form a tetra-gram pattern called P-Trigram, where the last cell is shown as a simplification with the next consecutive function, taken from the second trigram (e.g. trigram [IV·I·V] paired with [I·V·vi]), becomes p-trigram [IV·I·V → vi]. The gradient of black and white shows a stronger prevalence in darker cells. The figure on the left is sorted using Suno as the reference. The right figure uses Udio as a reference.

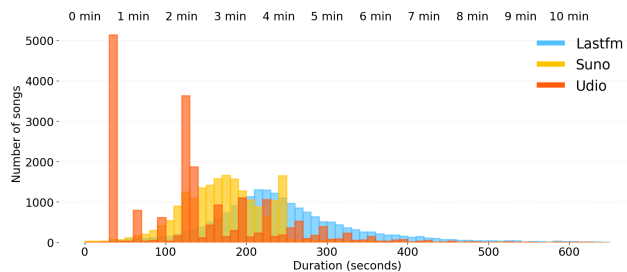


Figure 10: Distribution of song durations across the three collections (Lastfm, Suno, Udio). The x-axis represents duration in seconds, and the y-axis indicates the number of songs per 10-second bin.

the $1.3 \times - 2.1 \times$ range. The cumulative effect across all 53 enriched progressions is substantial: Suno’s harmonic vocabulary is contracted to 6,770 unique trigram types, compared to 8,056 in Udio and 9,838 in Lastfm.

The Axis of Awesome analysis extends this picture to the four-chord level. At the $\geq 10\%$ axis p-trigram threshold, 32.0% of Suno songs qualify as axis-based, versus 12.6% in Lastfm and 14.7% in Udio (Table 5). At the strictest level ($\geq 50\%$ of harmonic windows matching the Axis loop), Suno reaches 10.5% of songs, approximately $4.6 \times$ the rate observed in Lastfm (2.3%) and $3.5 \times$ that of Udio (3.0%). Songs built on canonical axis-family patterns are thus present at roughly 2.5 to 4.6 times the rate found in the human reference corpus, depending on the strictness of the threshold applied.

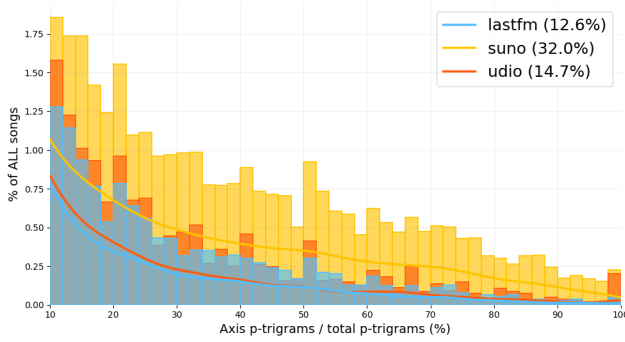


Figure 11: Distribution of the percentage of axis-based p-trigrams per song across the three collections (songs with $\geq 10\%$ axis content only). Lastfm shows 12.6%, Suno 32.0%, and Udio 14.7% of songs exceeding the 10% axis p-trigram threshold.

Udio’s harmonic distributions, by contrast, are statistically closer to those of Lastfm across all three measures. Cross-collection trigram frequency correlations confirm this: Udio and Lastfm share $r = 0.952$, while Suno correlates with Lastfm at only $r = 0.441$. The BPE merge histories of harmonic sequences (Figures 9) reinforce this at the tetragram level, with Udio and Lastfm sharing similar merge-pair frequency profiles while Suno diverges toward a smaller set of high-frequency patterns. Udio features minor-key functional progressions (e.g., $bVI-bVII-i$, $i-bVI-III$, $i-bVII-bVI$) among its most frequent trigrams (see Figure 2). Progressions of this type, particularly the $bVI-bVII-i$ pattern, have been identified as characteristic harmonic idioms of rock music, functioning as Aeolian cadences with subdominant–dominant–tonic implications distinct from conventional major–minor tonality (Biamonte, 2010). Lastfm’s top trigrams are dominated by functional major progressions ($IV-I-V$, $IV-V-I$, $V-IV-I$), with minor-mode patterns present but appearing further down the frequency ranking (see Figure 2).

6.3 RQ3: Structural Similarities and Differences

The BPE analysis of section-label sequences and the Unique Tokens Ratio provide complementary views of formal diversity across the three collections.

Suno exhibits the strongest convergence toward simple, repetitive structures. Verse-chorus cycles (ABAB patterns) dominate its merge history, and its Unique Tokens Ratio is the lowest of the three collections (0.44 ± 0.18), indicating that the average Suno track draws from fewer distinct section types than either human-composed or Udio-generated music. Udio, by contrast, shows the highest formal diversity (0.58 ± 0.20), with Lastfm occupying an intermediate position (0.51 ± 0.19). The BPE merge-pair heatmap (Figure 7) confirms that Udio’s merge frequencies are closer to Lastfm than to Suno. Of particular note, the second most frequent pattern in both AI collections

— $verse \cdot chorus \cdot inst \rightarrow verse$ (Suno: 2,103; Udio: 1,599) — is entirely absent from Lastfm. Both systems frequently produce instrumental sections as structural connectors within verse-chorus cycles, a formal strategy with no parallel in the human corpus.

The normalized musical form trigram profiles per genre (Figure 8) reveal a consistent pattern across styles. Both Suno and Udio show a reduced range of formal options relative to Lastfm, with intro-verse-chorus sequences markedly under-represented and solo sections largely absent. This is consistent with how both platforms operate: users generate a song segment that begins directly with a verse-chorus cycle and can later be extended via the web interface. The result is that intro sections, which precede the core structure, and solo sections, which typically appear later in human-composed music, are rarely generated in the initial output. The same genre-ontology caveat raised for Figure 5 applies here: the Jazz polygon in Figure 8 should not be read as a direct cross-collection comparison of jazz formal conventions, since the tracks grouped under that label differ substantially in character across the three corpora.

A key confound is song duration. Suno tracks average 175 seconds (approximately 2.9 minutes) and Udio tracks average 143 seconds (approximately 2.4 minutes), compared to 261 seconds (approximately 4.4 minutes) for Lastfm (Figure 10). Shorter tracks have fewer opportunities for structural elaboration (bridges, instrumental breaks, key changes), so the reduced diversity observed in Suno and Udio may partly reflect a generation-length constraint rather than a purely compositional bias. Udio’s duration distribution is notably bimodal: a sharp spike below one minute reflects tracks published at the platform’s default clip length of 32 seconds without extension, while the secondary peak near 130 seconds corresponds to tracks generated at the platform’s longest available clip length (2 minutes and 10 seconds).

A further caveat concerns formal labels and musical function. The All-In-One model assigns labels based on acoustic properties, but a verse-chorus alternation in a generated track may represent static loop repetition rather than the narrative-arc form those labels imply in conventional songwriting (Nieto et al., 2020). Distinguishing conventional pop form from non-teleological loop structures would require additional features such as melodic contour, rhythmic analysis, or lyric structure.

6.4 Outcome Homogenization and Creative Agency

The harmonic compression documented in RQ2 and the structural narrowing documented in RQ3 both point toward a form of outcome homogenization, consistent with the concept of algorithmic monoculture, where shared components across AI systems increase the likelihood of convergent outputs (Bommasani et al., 2022). Suno exemplifies this pattern, converging on a statisti-

cal peak of canonical pop progressions (Figure 6, Table 4), also visible in the Byte-Pair Encoding of the most common p-trigrams (Figure 9). The output does not reflect the variety of harmonic practice or formal organization found in human-composed music. Several mechanisms may contribute to this convergence. First, training data bias amplification, in which generative models learn to reproduce only the most frequent patterns in their training corpora. Since contemporary pop music dominates streaming platforms and likely constitutes a significant portion of Suno’s training data, these harmonic patterns are overrepresented in its outputs. Second, optimization toward perceptual plausibility encourages models to generate music that sounds “familiar” to broad audiences, reinforcing existing harmonic conventions rather than exploring innovative progressions. Third, the lack of harmonic-prompting capabilities in current platforms means that users cannot directly specify harmonic content, instead relying on the models’ learned biases toward common progressions. Fourth, Suno’s developers may intentionally be targeting users who prefer a narrower spectrum of music, focusing on commercial modern pop styles. Our data cannot determine which of these factors is primary, but the contrast with Udio is informative. Udio produces more diverse harmonic and structural outputs than would be expected from a presumably comparable training distribution, suggesting that model architecture and generation strategy play a role beyond the training data alone. These platforms promise to make music creation widely accessible, but generating convincing audio recordings does not necessarily entail preserving musical knowledge or variety. We note that a restricted harmonic vocabulary is not in itself a compositional deficiency: twelve-bar blues, punk, and minimal techno are all built on narrow chord sets by deliberate stylistic choice. Our claim is different: the data show that Suno applies a single harmonic template across all stated genres, a platform-level convergence that is independent of user intent. This is consistent with Suno’s own positioning: as documented by Neely (2024), the platform’s leadership has described it explicitly as gamifying musical creation. This raises questions about whether these tools genuinely support creative agency among musicians with any degree of musical knowledge (Wiggins, 2007, 2016; Lin et al., 2023; Tan, 2024). The prevalence of simple, repetitive formal structures (as revealed by our BPE analysis shown in Figure 9) further suggests that current platforms may be optimized to generate familiar, commercially viable content rather than to support experimentation or skill acquisition. This optimization strategy may serve commercial interests, but it limits the creative potential of AI as a compositional tool.

7. Conclusion

Our analysis of 60,014 tracks reveals systematic differences between AI-generated and human-composed popular music. Suno converges on a narrow harmonic and formal space: canonical pop progressions are overrepresented at 2.5 to 4.6 times the rate found in human-composed music, and its structural repertoire is dominated by simple verse-chorus cycles. Udio’s harmonic and structural distributions are statistically closer to those of human-composed music across all measures examined. These findings, and their implications for outcome homogenization and creative agency, are discussed in detail in Section 6. One direction for future work follows directly from the limitations identified in Section 6.1: improving functional harmonic labeling to handle modulations, modal mixture, and key ambiguity would strengthen corpus-scale analyses that currently depend on a single global key estimate. More broadly, this work demonstrates that fully automated, open-source MIR pipelines can now support corpus-scale harmonic and formal analysis at a scale and vocabulary richness that were not previously practical. The combination of rich-vocabulary chord estimation and semantic section labeling, applied here to 60,014 tracks, offers a replicable methodology for computational musicology and MIR as AI-generated music continues to grow in both volume and diversity.

8. Reproducibility

We release the complete dataset (60,014 JSON files with beat-level chord annotations, segmentation boundaries, style metadata, and audio links) and all analysis code under an MIT license.¹² GitHub repository.

9. Ethics and consent

Although the full ramifications of AI-generated material remain uncertain, the sheer speed and volume at which these systems can create new works demand scholarly attention (Sturm et al., 2024; Pelly, 2025). Because our study centers on music, we gathered examples from the commercial services Suno and Udio. Doing so raises a set of ethical issues. Even when a site’s terms of use do not appear to permit automated collection, the legitimacy of repurposing that material for research—particularly under a “fair use” rationale—deserves scrutiny. As a publicly funded team investigating AI music and its broader consequences, we rely on the text-and-data-mining exception laid out in the EU Copyright Directive 2019/790 (European Union, 2019) to justify our data acquisition.

Notes

¹ <https://suno.com>, last accessed Sep. 30 2025.

² <http://bit.ly/4nnZA55>, last accessed Sep. 30 2025.

³ <https://www.udio.com>, last accessed Sep. 30 2025.

⁴ <https://bit.ly/46Q0Gyy>, last accessed Sep. 30 2025.

⁵ <https://www.theguardian.com/technology/2025/jul/14/an-ai-generated-band-got-1m-plays-on-spotify-now-music-insiders-say-listeners-should-be-warned>, last accessed Sep. 30 2025.

⁶ <https://www.musicbusinessworldwide.com/nearly-a-third-of-all-tracks-uploaded-to-deezer-are-now-fully-ai-generated-says-platform/>, last accessed Sep. 30 2025.

⁷ https://anonymous.4open.science/r/Aimir_Structure_Analysis

⁸ <http://isophonics.net/content/reference-annotations>, last accessed Sep. 30 2025.

⁹ <https://suno.com/playlist/cc14084a-2622-4c4b-8258-1f6b4b4f54b3>, last accessed Sep. 30, 2025.

¹⁰ https://github.com/renesemela/lastfm-dataset-2020/raw/master/datasets/lastfm_dataset_2020/lastfm_dataset_2020.db, last accessed Sep. 30 2025.

¹¹ We used yt-dlp <https://github.com/yt-dlp/yt-dlp>, last accessed Sep. 30 2025.

¹² theproject

Acknowledgements

Contributions

XXX did ... YYY did ...

References

- Agostinelli, A., Denk, T. I., Borsos, Z., Engel, J., Verzetzi, M., Caillon, A., Huang, Q., Jansen, A., Roberts, A., Tagliasacchi, M., et al. (2023). Musi-clm: Generating music from text. arXiv preprint arXiv:2301.11325.
- Ariza, C. (2009). The interrogator as critic: The Turing test and the evaluation of generative music systems. *Computer Music Journal*, 33(2):48–70.
- Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal statistical society: series B (Methodological)*, 57(1):289–300.
- Bertin-Mahieux, T., Ellis, D. P., Whitman, B., and Lamere, P. (2011). The million song dataset. In *ISMIR*, pages 591–596.
- Biamonte, N. (2010). Triadic modal and pentatonic patterns in rock music. *Music Theory Spectrum*, 32(2):95–110.
- Böck, S., Korzeniowski, F., Schlüter, J., Krebs, F., and Widmer, G. (2016). Madmom: A new python audio and music signal processing library. In *Proc. of the 24th ACM International Conference on Multimedia*, pages 1174–1178. ACM.
- Bommasani, R., Creel, K. A., Kumar, A., Jurafsky, D., and Liang, P. S. (2022). Picking on the same

person: Does algorithmic monoculture lead to outcome homogenization? *Advances in Neural Information Processing Systems*, 35:3663–3678.

Buisson, M., Mcfee, B., and Essid, S. (2024). Using pairwise link prediction and graph attention networks for music structure analysis. In *ISMIR*.

Buisson, M., Mcfee, B., Essid, S., and Crayencour, H.-C. (2022). Learning multi-level representations for hierarchical music structure analysis. In *ISMIR*.

Burgoyne, J. A., Wild, J., and Fujinaga, I. (2011). An expert ground truth set for audio chord recognition and music analysis. In *ISMIR*, volume 11, pages 633–638.

Casini, L., Vila, L. C., Dalmazzo, D., Kaila, A.-K., and Sturm, B. L. (2025). Data-driven analysis of text-conditioned ai-generated music: A case study with suno and udio. *TISMIR*.

Collins, N. (2025). Recording artist career comparison through audio content analysis. *Royal Society Open Science*, 12(7):241647.

Copet, J., Kreuk, F., Gat, I., Remez, T., Kant, D., Synnaeve, G., Adi, Y., and Défossez, A. (2023). Simple and controllable music generation. *Advances in Neural Information Processing Systems*, 36:47704–47720.

Cros Vila, L., Sturm, B., Casini, L., and Dalmazzo, D. (2025). The ai music arms race: On the detection of ai-generated music. *Transactions of the International Society for Music Information Retrieval*, 8(1):179–194.

Cuthbert, M. S. and Ariza, C. (2010). music21: A toolkit for computer-aided musicology and symbolic music data. In *Proc. of the 11th International Society for Music Information Retrieval Conference*, pages 637–642.

Déguernel, K. and Sturm, B. L. T. (2023). Bias in favour or against computational creativity: A survey and reflection on the importance of socio-cultural context in its evaluation. In *Proc. Int. Conf. Comp. Creativity*.

Dervakos, E., Filandrianos, G., and Stamou, G. (2021). Heuristics for evaluation of ai generated music. In *2020 25th international conference on pattern recognition (ICPR)*, pages 9164–9171. IEEE.

European Union (2019). Directive (EU) 2019/790 of the European Parliament and of the Council of 17 April 2019 on copyright and related rights in the Digital Single Market. *Official Journal of the European Union*, L 130, 17 May 2019, pp. 92–125.

Evans, Z., Parker, J. D., Carr, C., Zukowski, Z., Taylor, J., and Pons, J. (2025). Stable audio open. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE.

Foote, J. (2000). Automatic audio segmentation using a measure of audio novelty. In *2000 IEEE international conference on multimedia and expo*.

- icme2000. proceedings. latest advances in the fast changing world of multimedia (cat. no. 00th8532), volume 1, pages 452–455. IEEE.
- Humphrey, E. J. and Bello, J. P. (2015). Four timely insights on automatic chord estimation. In *ISMIR*, volume 10, pages 673–679.
- Jiang, J., Chen, K., Li, W., and Xia, G. (2019). Large-vocabulary chord transcription via chord structure decomposition. In *ISMIR*, pages 644–651.
- Kim, T. and Nam, J. (2023). All-in-one metrical and functional structure analysis with neighborhood attentions on demixed audio. In *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*.
- Krumhansl, C. L. and Kessler, E. J. (1982). Tracing the dynamic changes in perceived tonal organization in a spatial representation of musical keys. *Psychological review*, 89(4):334.
- Levy, M. and Sandler, M. (2008). Structural segmentation of musical audio by constrained clustering. *IEEE transactions on audio, speech, and language processing*, 16(2):318–326.
- Lin, Z., Ehsan, U., Agarwal, R., Dani, S., Vashishth, V., and Riedl, M. (2023). Beyond prompts: Exploring the design space of mixed-initiative co-creativity systems.
- Marmoret, A., Cohen, J. E., and Bimbot, F. (2023). Barwise music structure analysis with the correlation block-matching segmentation algorithm. *TISMIR*, 6(1):167–185.
- Mauch, M. and Dixon, S. (2009). Simultaneous estimation of chords and musical context from audio. *IEEE Transactions on Audio, Speech, and Language Processing*, 18(6):1280–1289.
- Mauch, M. and Dixon, S. (2010). Approximate note transcription for the improved identification of difficult chords. In *ISMIR*, pages 135–140.
- Mauch, M., MacCallum, R. M., Levy, M., and Leroi, A. M. (2015). The evolution of popular music: Usa 1960–2010. *Royal Society open science*, 2(5):150081.
- McFee, B. and Bello, J. P. (2017). Structured training for large-vocabulary chord recognition. In *ISMIR*, pages 188–194.
- McFee, B. and Ellis, D. P. (2014). Learning to segment songs with ordinal linear discriminant analysis. In *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5197–5201. IEEE.
- McFee, B., Raffel, C., Liang, D., Ellis, D. P., McVicar, M., Battenberg, E., and Nieto, O. (2025). *librosa*.
- Müller, M. (2015). *Fundamentals of Music Processing: Audio, Analysis, Algorithms, Applications*. Springer.
- Neely, A. (2024). [video title]. YouTube, accessed [21-05-2026].
- Ni, Y., McVicar, M., Santos-Rodriguez, R., and De Bie, T. (2012). An end-to-end machine learning system for harmonic analysis of music. *IEEE Transactions on Audio, Speech, and Language Processing*, 20(6):1771–1783.
- Nieto, O., Mysore, G. J., Wang, C.-i., Smith, J. B., Schlüter, J., Grill, T., and McFee, B. (2020). Audio-based music structure analysis: Current trends, open challenges, and applications. *Transactions of the International Society for Music Information Retrieval*, 3(1).
- Odekerken, D., Koops, H. V., and Volk, A. (2020). Decibel: Improving audio chord estimation for popular music by alignment and integration of crowd-sourced symbolic representations. *arXiv preprint arXiv:2002.09748*.
- Oramas, S., Barbieri, F., Nieto, O., and Serrà, X. (2018). Multimodal deep learning for music genre classification. *Transactions of the International Society for Music Information Retrieval*, 1(1):4–22.
- Park, J., Choi, K., Jeon, S., Kim, D., and Park, J. (2019). A bi-directional transformer for musical chord recognition.
- Pasquier, P., Burnett, A., Thomas, N. G., Maxwell, J. B., Eigenfeldt, A., and Loughin, T. (2016). Investigating listener bias against musical metacreativity. In *Proc. ICCM*.
- Pauwels, J. and Peeters, G. (2013). Evaluating automatically estimated chord sequences. In *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 749–753. IEEE.
- Peeters, G. (2023). Self-similarity-based and novelty-based loss for music structure analysis. *ISMIR*.
- Pelly, L. (2025). *Mood machine: The rise of Spotify and the costs of the perfect playlist*. Simon and Schuster.
- Poltronieri, A., Serrà, X., and Rocamora, M. (2025). From discord to harmony: Decomposed consonance-based training for improved audio chord estimation. In *ISMIR. International Society for Music Information Retrieval, ISMIR 2025*.
- Poppe, T., Lopes, L., and Figueiredo, F. (2025). The polyphonic audio to roman corpus. In *Proceedings of the 12th International Conference on Digital Libraries for Musicology*, pages 72–80.
- Ragot, M., Martin, N., and Cojean, S. (2020). Ai-generated vs. human artworks. a perception bias towards artificial intelligence? In *Proc. CHI*.
- Richards, M. (2017). Tonal ambiguity in popular music’s axis progressions. *Music Theory Online*, 23(3).
- Rouard, S., Massa, F., and Défossez, A. (2023). Hybrid transformers for music source separation. In *ICASSP 2023-2023 IEEE International Confer-*

- ence on Acoustics, Speech and Signal Processing (ICASSP), pages 1–5. IEEE.
- Salamon, J., Nieto, O., and Bryan, N. J. (2021). Deep embeddings and section fusion improve music segmentation. In ISMIR, pages 594–601.
- Sennrich, R., Haddow, B., and Birch, A. (2016). Neural machine translation of rare words with subword units. In Proceedings of the 54th annual meeting of the association for computational linguistics (volume 1: long papers), pages 1715–1725.
- Serrà, J., Corral, Á., Boguñá, M., Haro, M., and Arcos, J. L. (2012). Measuring the evolution of contemporary western popular music. *Scientific reports*, 2(1):1–6.
- Serrà, J., Müller, M., Grosche, P., and Arcos, J. L. (2012). Unsupervised detection of music boundaries by time series structure features. In Proc. of the AAAI Conference on Artificial Intelligence, volume 26, pages 1613–1619.
- Shank, D. B., Stefanik, C., Stuhlsatz, C., Kacirek, K., and Belfi, A. M. (2022). Ai composer bias: Listeners like music less when they think it was composed by an ai. *J. Experimental Psychology: Applied*.
- Sturm, B. L. and Ben-Tal, O. (2017). Taking the models back to music practice: Evaluating generative transcription models built using deep learning. *J. Creative Music Systems*, 2(1).
- Sturm, B. L. T., Déguernel, K., Huang, R. S., Kaila, A.-K., Jääskeläinen, P., Kanhov, E., Cros Vila, L., Dalmazzo, D., Casini, L., Bown, O. R., Collins, N., Drott, E., Sterne, J., Holzapfel, A., and Ben-Tal, O. (2024). AI Music Studies: Preparing for the Coming Flood. In AIMC 2024, Oxford, United Kingdom.
- Tan, S. (2024). Are we all musicians now? authenticity, musicianship, and ai music generator suno.
- Temperley, D. and Clercq, T. d. (2013). Statistical analysis of harmony and melody in rock music. *Journal of New Music Research*, 42(3):187–204.
- Wang, J.-C., Hung, Y.-N., and Smith, J. B. (2022). To catch a chorus, verse, intro, or anything else: Analyzing a song with structural functions. In ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pages 416–420. IEEE.
- Waseem Akram, M., Dettori, S., Colla, V., and Carlo Buttazzo, G. (2025). Chordformer: A conformer-based architecture for large-vocabulary audio chord recognition. *IEEE TASLP*.
- Wiggins, J. (2007). Compositional process in music. In *International handbook of research in arts education*, pages 453–476. Springer.
- Wiggins, J. (2016). Musical agency. The child as musician: A handbook of musical development, pages 102–121.
- Wu, S.-L. and Yang, Y.-H. (2020). The jazz transformer on the front line: Exploring the shortcomings of ai-composed music through quantitative measures. *ISMIR*.
- Zlatkov, D., Ens, J., and Pasquier, P. (2023). Searching for human bias against ai-composed music. In Proc. EvoMUSART.